

```
-- ASSIGNMENT 8: EMPLOYEE MANAGEMENT SYSTEM
-- IN, BETWEEN, LIKE, ORDER BY, GROUP BY, HAVING
-- AND AGGREGATE FUNCTIONS
```

```
-- Main Schema
```

```
CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Email VARCHAR(100),
    Phone CHAR(10),
    Salary DECIMAL(10, 2),
    Gender CHAR(1),
    DepartmentID INT
);
```

```
-- Insert Records
```

```
INSERT INTO Employees VALUES
(101, 'Amit', 'Verma', 'amit.v@example.com', '9876543210', 55000.00, 'M', 1),
(102, 'Priya', 'Mehra', 'priya.m@example.com', '9123456789', 65000.00, 'F', 2),
(103, 'Ravi', 'Singh', 'ravi.s@example.com', '9988776655', 40000.00, 'M', 1),
(104, 'Sneha', 'Roy', 'sneha.r@example.com', '9876501234', 75000.00, 'F', 3),
(105, 'Anil', 'Kumar', 'anil.k@example.com', '9812345678', 30000.00, 'M', 2);
```

```
-- 1. Using IN Clause
```

```
INSECT *
FROM Employees
WHERE DepartmentID IN (1, 3);
```

```
-- 2. Using BETWEEN Clause
```

```
SELECT *
FROM Employees
WHERE Salary BETWEEN 40000 AND 70000;
```

```
-- 3. Using LIKE Clause
```

```
SELECT *
FROM Employees
WHERE FirstName LIKE 'A%';
```

```
-- 4. Using ORDER BY Clause
```

```
SELECT *
FROM Employees
ORDER BY Salary ASC;
```

```
-- 5. Using GROUP BY Clause
```

```
SELECT DepartmentID, COUNT(*) AS TotalEmployees
FROM Employees
GROUP BY DepartmentID;
```

```
-- 6. Using HAVING Clause
```

```
SELECT DepartmentID, COUNT(*) AS TotalEmployees
FROM Employees
GROUP BY DepartmentID
HAVING COUNT(*) > 1;
```

```
-- 7. Find the total number of employees in the department.
```

```
SELECT COUNT(*) AS TotalEmployees
FROM Employees;
```

```
-- 8. Find the total salary of all employees.
```

```
SELECT SUM(Salary) AS TotalSalary
FROM Employees;
```

```
-- 9. Find the average salary of all employees.
```

```
SELECT AVG(Salary) AS AverageSalary
FROM Employees;
```

```
-- 10. Find the highest salary of the employee.
```

```
SELECT MAX(Salary) AS HighestSalary
FROM Employees;
```

```
-- 11. Find the lowest salary of the employee.
```

```
SELECT MIN(Salary) AS LowestSalary
FROM Employees;
```

```
-- 12. Find the number of employees in each department.
```

```
SELECT DepartmentID, COUNT(*) AS TotalEmployees
FROM Employees
GROUP BY DepartmentID;
```

```
-- 13. Find the average salary of employees in each department.
```

```
SELECT DepartmentID, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY DepartmentID;
```

```
-- 14. Find the total salary paid by each department.
```

```
SELECT DepartmentID, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY DepartmentID;
```

```
-- 15. Count the number of male and female employees.
```

```
SELECT Gender, COUNT(*) AS TotalEmployees
FROM Employees
GROUP BY Gender;
```

```
-- 16. Cond the average salary of male and female employees.
```

```
SELECT Gender, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY Gender;
```

4	07:03:32	CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50), Email VARCHAR(100), Phone CHAR(10), Salary INT, Gender CHAR(1), DepartmentID INT) 0 row(s) affected
5	07:03:32	INSERT INTO Employees VALUES (101, 'Amit', 'Verma', 'amit.v@example.com', '9876543210', 55000.00, 'M', 1), (102, 'Priya', 'Mehra', 'priya.m@example.com', '9123456789', 65000.00, 'F', 2), (103, 'Ravi', 'Singh', 'ravi.s@example.com', '9988776655', 40000.00, 'M', 1), (104, 'Sneha', 'Roy', 'sneha.r@example.com', '9876501234', 75000.00, 'F', 3) 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	07:03:32	SELECT * FROM Employees WHERE DepartmentID IN (1, 3) LIMIT 0, 1000 3 row(s) returned
7	07:03:32	SELECT * FROM Employees WHERE Salary BETWEEN 40000 AND 70000 LIMIT 0, 1000 3 row(s) returned
8	07:03:32	SELECT * FROM Employees WHERE FirstName LIKE 'A%' LIMIT 0, 1000 2 row(s) returned
9	07:03:32	SELECT * FROM Employees ORDER BY Salary ASC LIMIT 0, 1000 5 row(s) returned
10	07:03:32	SELECT DepartmentID, COUNT(*) AS TotalEmployees FROM Employees GROUP BY DepartmentID LIMIT 0, 1000 3 row(s) returned
11	07:03:32	SELECT DepartmentID, COUNT(*) AS TotalEmployees FROM Employees GROUP BY DepartmentID HAVING COUNT(*) > 1 LIMIT 0, 1000 2 row(s) returned
12	07:03:32	SELECT COUNT(*) AS TotalEmployees FROM Employees LIMIT 0, 1000 1 row(s) returned
13	07:03:32	SELECT SUM(Salary) AS TotalSalary FROM Employees LIMIT 0, 1000 1 row(s) returned
14	07:03:32	SELECT AVG(Salary) AS AverageSalary FROM Employees LIMIT 0, 1000 1 row(s) returned
15	07:03:32	SELECT MAX(Salary) AS HighestSalary FROM Employees LIMIT 0, 1000 1 row(s) returned
16	07:03:32	SELECT MIN(Salary) AS LowestSalary FROM Employees LIMIT 0, 1000 1 row(s) returned
17	07:03:32	SELECT DepartmentID, COUNT(*) AS TotalEmployees FROM Employees GROUP BY DepartmentID LIMIT 0, 1000 3 row(s) returned
18	07:03:32	SELECT DepartmentID, AVG(Salary) AS AverageSalary FROM Employees GROUP BY DepartmentID LIMIT 0, 1000 3 row(s) returned
19	07:03:33	SELECT DepartmentID, SUM(Salary) AS TotalSalary FROM Employees GROUP BY DepartmentID LIMIT 0, 1000 3 row(s) returned
20	07:03:33	SELECT Gender, COUNT(*) AS TotalEmployees FROM Employees GROUP BY Gender LIMIT 0, 1000 2 row(s) returned
21	07:03:33	SELECT Gender, AVG(Salary) AS AverageSalary FROM Employees GROUP BY Gender LIMIT 0, 1000 2 row(s) returned

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Email	Phone	Salary	Gender	DepartmentID
▶	101	Amit	Verma	amit.v@example.com	9876543210	55000.00	M	1
	103	Ravi	Singh	ravi.s@example.com	9988776655	40000.00	M	1
	104	Sneha	Roy	sneha.r@example.com	9876501234	75000.00	F	3
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

1 x

2

3

4

5

6

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8

9

10

11

12

13

14

15

16

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Email	Phone	Salary	Gender	DepartmentID
▶	101	Amit	Verma	amit.v@example.com	9876543210	55000.00	M	1
	102	Priya	Mehra	priya.m@example.com	9123456789	65000.00	F	2
	103	Ravi	Singh	ravi.s@example.com	9988776655	40000.00	M	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

1

2 ×

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11

12

13

14

15

16

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Email	Phone	Salary	Gender	DepartmentID
▶	101	Amit	Verma	amit.v@example.com	9876543210	55000.00	M	1
	105	Anil	Kumar	anil.k@example.com	9812345678	30000.00	M	2
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

1

2

3 ×

4

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11

12

13

14

15

16

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Email	Phone	Salary	Gender	DepartmentID
▶	105	Anil	Kumar	anil.k@example.com	9812345678	30000.00	M	2
	103	Ravi	Singh	ravi.s@example.com	9988776655	40000.00	M	1
	101	Amit	Verma	amit.v@example.com	9876543210	55000.00	M	1
	102	Priya	Mehra	priya.m@example.com	9123456789	65000.00	F	2
	104	Sneha	Roy	sneha.r@example.com	9876501234	75000.00	F	3

1

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16

Result Grid



Filter Rows:

	DepartmentID	TotalEmployees
▶	1	2
	2	2
	3	1

1

2

3

4

5 ×

6

7

Result Grid		 Filter Rows:
	DepartmentID	TotalEmployees
	1	2
	2	2

Result Grid		 Filter Rows:
	TotalEmployees	
	5	

Result Grid		
	TotalSalary	
▶	265000.00	

8

×

9

10

Result Grid			
	AverageSalary		
▶	53000.000000		

8

9 ×

10

Result Grid	
	HighestSalary
▶	75000.00

Result Grid		
	LowestSalary	
▶	30000.00	

11 × 12 1

Result Grid			 Filter Rows:
	DepartmentID	TotalEmployees	
	1	2	
	2	2	
	3	1	

Result Grid			 Filter Rows:
	DepartmentID	AverageSalary	
	1	47500.000000	
	2	47500.000000	
	3	75000.000000	

13

×

14

15

16

Result Grid			 Filter Rows:
	DepartmentID	TotalSalary	
▶	1	95000.00	
	2	95000.00	
	3	75000.00	

14 × 15 16

Result Grid

Filter Rows:

	Gender	TotalEmployees
▶	M	3
	F	2

15 ×

16

Result Grid			 Filter Rows
	Gender	AverageSalary	
▶	M	41666.666667	
	F	70000.000000	

16 × 